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Release 2

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Contents

Intellectual Property Rights 5
Foreword 5
Modal verbs terminology 5
1 Scope 6
2 References 6
2.1 Normative references 6
2.2 Informative references 6
3 Definition of terms, symbols and abbreviations 7
3.1 Terms 7
3.2 Symbols 7
3.3 Abbreviations 7
4 Test Suite Structure (TSS) 8
4.1 Structure for Security tests 8
5 Test Purposes (TP) 8
5.1 Introduction 8
5.1.1 TP definition conventions 8
5.1.2 TP Identifier naming conventions 8
5.1.3 Rules for the behaviour description 9
5.1.4 Sources of TP definitions 9
5.1.5 Mnemonics for PICS reference 9
6 ITS-S Security 10
6.1 Overview 10
6.1.1 Certificates content 10
6.1.1.1 Root Certificate Authorities certificates 10
6.1.1.2 Authorization Authorities certificates 11
6.1.1.3 Authorization Tickets 12
6.2 Sending behaviour 14
6.2.1 General sending behaviour 14
6.2.1.1 Check the message protocol version 14
6.2.2 CAM profile 14
6.2.2.1 Check that secured CAM is signed 14
6.2.2.2 Check secured CAM AID value 15
6.2.2.3 Check header fields 15
6.2.2.4 Check signer information 16
6.2.2.5 Check that IUT sends certificate to unknown ITS-S 18
6.2.2.6 Check that IUT restarts the timer when the certificate has been sent 19
6.2.2.7 Check sending certificate request for unknown certificate 19
6.2.2.8 Check that IUT sends AT certificate when requested 21
6.2.2.9 Check that IUT sends AA certificate when requested 22
6.2.2.10 Check generation time 26
6.2.2.11 Check payload 26
6.2.2.12 Check signing permissions 27
6.2.2.13 Check signature 27
6.2.2.14 Check support for certificate content 28
6.2.2.15 Check certificate consistency conditions 29
6.2.3 DENM profile 31
6.2.3.1 Check secured DENM is signed 31
6.2.3.2 Check secured DENM AID value 31
6.2.3.3 Check header fields 32
6.2.3.4 Check signer information 32
6.2.3.5 Check generation time 33
6.2.3.6 Check generation location 33
6.2.3.7 Check payload 35

6.2.3.8 Check signing permissions 36
6.2.3.9 Check signature 36
6.2.3.10 Check support for certificate content 37
6.2.3.11 Check certificate consistency conditions 38
6.2.4 Generic signed message profile 40
6.2.4.1 Check that secured message is signed 40
6.2.4.2 Check secured AID value 40
6.2.4.3 Check header field 41
6.2.4.4 Check that signer info is a certificate or digest 41
6.2.4.5 Check generation time 42
6.2.4.6 Check payload 42
6.2.4.7 Check signing permissions 43
6.2.4.8 Check signature 43
6.3 Receiving behaviour 44
6.3.1 Check the message protocol version 44
6.3.2 CAM profile 45
6.3.2.1 Check the valid message receiving 45
6.3.2.2 Check invalid HeaderInfo elements 48
6.3.2.3 Check invalid Signature elements 50
6.3.3 DENM profile 51
6.3.3.1 Check the valid message receiving 51
6.3.3.2 Check invalid HeaderInfo elements 54
6.3.3.3 Check invalid Signature elements 56
6.4 Implicit certificate 57

Annex A (informative): Bibliography 59

History 60

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Intelligent Transport Systems (ITS).

The present document is part 2 of a multi-part deliverable. Full details of the entire series can be found in part 1 [3].

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document provides the Test Suite Structure and Test Purposes (TSS & TP) for Security as defined in ETSI TS 103 097 [1] in accordance with the relevant guidance given in ISO/IEC 9646-7 [i.6].

The ISO standards for the methodology of conformance testing (ISO/IEC 9646-1 [i.3] and ISO/IEC 9646-2 [i.4]) as well as the ETSI rules for conformance testing (ETSI ETS 300 406 [i.7]) are used as a basis for the test methodology.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference/.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 103 097 (2.1.1): "Intelligent Transport Systems (ITS); Security; Security header and certificate formats; Release 2".

[2] IEEE Std 1609.2™-2016 including amendments IEEE Std 1609.2a™-2017 and IEEE Std 1609.2b™-2019: "IEEE Standard for Wireless Access in Vehicular Environments -- Security Services for Applications and Management Messages".

[3] ETSI TS 103 096-1 (V2.1.1): "Intelligent Transport Systems (ITS); Testing; Conformance test specifications for ITS Security; Part 1: Protocol Implementation Conformance Statement (PICS); Release 2".

[4] ETSI TS 102 871-1 (V1.5.1): "Intelligent Transport Systems (ITS); Testing; Conformance test specifications for GeoNetworking ITS-G5; Part 1: Test requirements and Protocol Implementation Conformance Statement (PICS) pro forma".

[5] Void.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI EG 202 798 (V1.1.1): "Intelligent Transport Systems (ITS); Testing; Framework for conformance and interoperability testing".

[i.2] ETSI TS 102 965 (V2.2.1): "Intelligent Transport Systems (ITS); Application Object Identifier (ITS-AID); Registration; Release 2".

[i.3] ISO/IEC 9646-1 (1994): "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 1: General concepts".

[i.4] ISO/IEC 9646-2 (1994): "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 2: Abstract Test Suite specification".

[i.5] ISO/IEC 9646-6 (1994): "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 6: Protocol profile test specification".

[i.6] ISO/IEC 9646-7 (1995): "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 7: Implementation Conformance Statements".

[i.7] ETSI ETS 300 406 (1995): "Methods for testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TS 103 097 [1], ETSI TS 102 965 [i.2], ISO/IEC 9646-6 [i.5] and ISO/IEC 9646-7 [i.6] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AA	Authorization Authority
AID	Application Identifier
AID_CAM	ITS Application Identifier for CAM
AID_DENM	Application Identifier for DENM
AID_GN	Application Identifier for general GeoNetworking messages
AT	Authorization Ticket
ATS	Abstract Test Suite
BO	Exceptional Behaviour
BV	Valid Behaviour
CA	Certificate Authority
CAM	Co-operative Awareness Messages
CAN	Controller Area Network
CERT	Certificate
DE	Data Element
DEN	Decentralized Environmental Notification
DENM	Decentralized Environmental Notification Message
EA	Enrolment Authority
ECC	Elliptic Curve Cryptography
GN	GeoNetworking
ITS	Intelligent Transport Systems
ITS-S	Intelligent Transport System - Station
IUT	Implementation under Test
MSG	Message
PICS	Protocol Implementation Conformance Statement
PSID	Provider Service Identifier
RCA	Root Certificate Authority
SSP	Service Specific Permissions

TP Test Purposes
TSS Test Suite Structure

4 Test Suite Structure (TSS)

4.1 Structure for Security tests

Table 1 shows the Security Test Suite Structure (TSS) defined for conformance testing.

Table 1: TSS for Security

Root	Group	Category
Security	ITS-S data transfer	Valid
	ITS-S - AA authorization	Valid
	ITS-S - EA enrolment	Valid
	Sending behaviour	Valid
	Receiving behaviour	Valid and Invalid
	Generic messages	Valid
	CAM testing	Valid
	DENM testing	Valid

5 Test Purposes (TP)

5.1 Introduction

5.1.1 TP definition conventions

The TP definition is built according to ETSI EG 202 798 [i.1].

5.1.2 TP Identifier naming conventions

The identifier of the TP is built according to table 2.

Table 2: TP naming convention

Identifier	TP_<root>_<tgt>_<gr>_<sgr>_<rn>_<sn>_<x>[_<v>]		
	<root> = root	SEC	
	<tgt> = target	ITSS	ITS-S data transfer
		CA	Certificate Authority tests
		AA	ITS-S - AA authorization
		EA	ITS-S - EA enrolment
	<gr> = group	SND	Sending behaviour
		RCV	Receiving behaviour
	<sgr> =sub- group	MSG	Generic messages
		CAM	CAM testing
		DENM	DENM testing
	<sn> = test purpose sequential number		01 to 99
	<x> = category	BV	Valid Behaviour tests
		BO	Invalid Behaviour Tests
	<v> = variant (optional)		A to Z

5.1.3 Rules for the behaviour description

The description of the TP is built according to ETSI EG 202 798 [i.1].

ETSI TS 103 097 [1] does not use the finite state machine concept. As a consequence, the test purposes use a generic "Initial State" that corresponds to a state where the IUT is ready for starting the test execution. Furthermore, the IUT shall be left in this "Initial State", when the test is completed.

Being in the "Initial State" refers to the starting point of the initial device configuration. There are no pending actions, no instantiated buffers or variables, which could disturb the execution of a test.

5.1.4 Sources of TP definitions

All TPs have been specified according to ETSI TS 103 097 [1] and IEEE Std 1609.2 [2].

5.1.5 Mnemonics for PICS reference

To avoid an update of all TPs when the PICS document is changed, table 3 introduces mnemonics name and the correspondence with the real PICS item number. The 'PICS item' as defined in IEEE Std 1609.2 [2], ETSI TS 103 096-1 [3] and ETSI TS 102 871-1 [4] shall be used to determine the test applicability.

Table 3: Mnemonics for PICS reference

	Mnemonic	PICS item
1	PICS_GN_SECURITY	A.2/1 [4]
2	PICS_SEC_CERTIFICATE_SELECTION	A.8/1 [3]
3	PICS_SEC_CIRCULAR_REGION	S1.2.2.5.1.1 [2]
4	PICS_SEC_RECTANGULAR_REGION	S1.2.2.5.1.2 [2]
5	PICS_SEC_POLYGONAL_REGION	S1.2.2.5.1.3 [2]
6	PICS_SEC_IDENTIFIED_REGION	S1.2.2.5.1.4 [2]
7	PICS_SEC_ITS_AID_OTHER	A.7/1 [3]
8	PICS_SEC_SHA256	S1.2.2.1.1 [2]
9	PICS_SEC_SHA384	S1.2.2.1.2 [2]
10	PICS_SEC_BRAINPOOL_P256R1	S1.2.2.4.1.2 [2]
11	PICS_SEC_BRAINPOOL_P384R1	S1.2.2.4.2 [2]
12	PICS_SEC_IMPLICIT_CERTIFICATE	S1.2.2.8 [2]
13	PICS_SEC_P2P_AT_DISTRIBUTION	S3 [2]
14	PICS_SEC_P2P_AA_DISTRIBUTION	S3 [2]

6 ITS-S Security

6.1 Overview

6.1.1 Certificates content

6.1.1.1 Root Certificate Authorities certificates

RCA certificate	Content	To be installed on the IUT
CERT_IUT_A_RCA	- self-signed - name "ETSI Test RCA A certificate" - application permissions: o CRL with SSP 0x01 o CTL with SSP 0x0138 - certificate issuing permissions: o CAM with all possible SPP (0x01FFFC / 0xFF0003) o DENM with all possible SSP (0x01FFFFFF / 0xFF000000) o SPATEM with all possible SSP (0x01E0 / 0xFF1F) o MAPEM with all possible SSP (0x01C0 / 0xFF3F) o IVIM with all possible SSP (0x01000000FFF8 / 0xFF0000000007) o SREM with all possible SSP (0x01FFFFE0 / 0xFF00001F) o SSEM with all possible SSP (0x01 / 0xFF) o GPC with all possible SSP (0x01 / 0xFF) o GN-MGMT without SSP o CRT-REQ with SSP (0x01FE / 0xFF01) - validation time for 3 years - no region restriction - assurance level 6 - verification key of type compressed with NIST P256R curve - valid signature of type x-only with NIST P256R curve	Yes
CERT_IUT_A_RCA_A8	Same as CERT_IUT_A_ATCERT_IUT_A_RCA, excepting the following: - certificate issuing permissions: o same as in CERT_IUT_A_RCA o unallocated ITS AIDs: 96, 97, 98, 99, 100, 101, 102 without SSP	Yes
CERT_IUT_C_RCA	Same as CERT_IUT_A_ATCERT_IUT_A_RCA, excepting the following: - rectangular region restriction (10 km square) - no unallocated ITS AID in certificate issuing permissions	Yes

6.1.1.2 Authorization Authorities certificates

AA certificate	Content	To be installed on the IUT
CERT_IUT_A_AA	- signer digest of the CERT_IUT_A_RCA - application permissions: o CRT_REQ with SSP 0x0132 - certificate issuing permissions: o CAM with all possible SPP (0x01FFFC / 0xFF0003) o DENM with all possible SSP (0x01FFFFFF / 0xFF000000) o SPATEM with all possible SSP (0x01E0 / 0xFF1F) o MAPEM with all possible SSP (0x01C0 / 0xFF3F) o IVIM with all possible SSP (0x01000000FFF8 / 0xFF0000000007) o SREM with all possible SSP (0x01FFFFE0 / 0xFF00001F) o SSEM with all possible SSP (0x01 / 0xFF) o GPC with all possible SSP (0x01 / 0xFF) o GN-MGMT without SSP - validation time for 3 years - no region restriction - assurance level 4 - verification key of type compressed with NIST P256R curve - encryption key of type compressed with NIST P256R curve - valid signature of type x-only with NIST P256R curve	Yes
CERT_IUT_A_N_AA	Same as CERT_IUT_A_ATCERT_IUT_A_AA, excepting the following: - verification key of type uncompressed	Yes
CERT_IUT_A_B_AA	Same as CERT_IUT_A_ATCERT_IUT_A_AA, excepting the following: - verification key with Brainpool P256r1 curve	Yes
CERT_IUT_A_B3_AA	Same as CERT_IUT_A_ATCERT_IUT_A_B_AA, excepting the following: - verification key with Brainpool P384r1 curve	Yes
CERT_IUT_A_AA_A8	Same as CERT_IUT_A_ATCERT_IUT_A_AA, excepting the following: - signer digest of the CERT_IUT_A_RCA_A8 - certificate issuing permissions: o CAM with all possible SPP (0x01FFFC / 0xFF0003) o unallocated ITS AIDs: 96, 97, 98, 99, 100, 101, 102 without SSP o no other certificate issuing permissions	Yes
CERT_IUT_CC_AA	Same as CERT_IUT_A_ATCERT_IUT_A_AA, excepting the following: - signer digest of the CERT_IUT_C_RCA - rectangular region restriction equal to the one in the CERT_IUT_C_RCA	Yes
CERT_IUT_C3_AA	Same as CERT_IUT_A_ATCERT_IUT_CC_AA, excepting the following: - rectangular region restriction oversizing the one in the CERT_IUT_C_RCA	Yes
CERT_IUT_CA_AA	Same as CERT_IUT_A_ATCERT_IUT_CC_AA, excepting the following: - no region restriction	Yes
CERT_IUT_D_AA	Same as CERT_IUT_A_ATCERT_IUT_CC_AA, excepting the following: - polygonal region restriction as a square with the side of 10 km and center in the IUT position	Yes
CERT_TS_A_AA	Same as CERT_IUT_A_ATCERT_IUT_A_AA. To be used on the Test System side.	Yes
CERT_TS_B_AA	Same as CERT_IUT_A_ATCERT_IUT_A_B_AA. To be used on the Test System side.	Yes
CERT_TS_A_B_AA	Same as CERT_IUT_A_ATCERT_IUT_A_B_AA. To be used on the Test System side.	Yes

6.1.1.3 Authorization Tickets

Authorization ticket	Content	To be installed on the IUT
CERT_IUT_A_AT	- Explicit certificate - signer digest of the CERT_IUT_A_AA; - application permissions: o CAM with all SPP (0x01FFFC); o DENM with all SSP (0x01FFFFFF); o GN-MGMT; - validation time for 1 year; - no region restriction; - assurance level 3; - verification key of type compressed with NIST P256R curve ; - encryption key of type compressed with NIST P256R curve; - valid signature of type x-only with NIST P256R curve.	Yes
CERT_IUT_A_AT_IMP	- Implicit certificate - signer digest of the CERT_IUT_A_AA; - application permissions: o CAM with all SPP (0x01FFFC); o DENM with all SSP (0x01FFFFFF); o GN-MGMT; - validation time for 1 year; - no region restriction; - assurance level 3; - reconstruction value of type compressed with NIST P256R curve ; - encryption key of type compressed with NIST P256R curve.	Yes
CERT_IUT_A_AT_IMP_BO	Same as CERT_IUT_A_AT_IMP, exception the following: - valid signature of type x-only with NIST P256R curve.	No
CERT_IUT_A_N_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - verification key of type uncompressed.	Yes
CERT_IUT_A_B_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - signer digest of the CERT_IUT_A_B_AA; - verification key with Brainpool P256r1 curve; - valid signature with Brainpool P256r1 curve.	Yes
CERT_IUT_A_B_N_AT	Same as CERT_IUT_A_ATCERT_IUT_A_B_AT, excepting the following: - verification key of type uncompressed.	Yes
CERT_IUT_A_B3_AT	Same as CERT_IUT_A_ATCERT_IUT_A_B_AT, excepting the following: - verification key with Brainpool P384r1 curve.	Yes
CERT_IUT_A_B3_N_AT	Same as CERT_IUT_A_ATCERT_IUT_A_B3_AT, excepting the following: - verification key of type uncompressed.	Yes

Authorization ticket	Content	To be installed on the IUT
CERT_IUT_A_B33_AT	Same as CERT_IUT_A_ATCERT_IUT_A_B3_AT, excepting the following: - signer digest of the CERT_IUT_A_B3_AA; - valid signature with Brainpool P384r1 curve.	Yes
CERT_IUT_A_AT_A8	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - signer digest of the CERT_IUT_A_AA_A8; - application permissions: o CAM with all SPP (0x01FFFC); o unallocated ITS AIDs: 96, 97, 98, 99, 100, 101, 102 without SSP.	Yes
CERT_IUT_B_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - circular region restriction with the radius of 5 km and center at the IUT point.	Yes
CERT_IUT_C_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - rectangular region restriction with the side of 5 km and center at the IUT point.	Yes
CERT_IUT_D_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - signer digest of the CERT_IUT_D_AA; - polygonal region restriction identical to the one in the CERT_IUT_D_AA, including the IUT position.	Yes
CERT_IUT_D_AT_8	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - polygonal region restriction contains 8 points.	Yes
CERT_IUT_E_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - identified region restriction including the IUT point.	Yes
CERT_IUT_E_AT_8	Same as CERT_IUT_A_ATCERT_IUT_E_AT, excepting the following: - identified region restriction contains 8 region identifiers.	Yes
CERT_IUT_A1_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - the certificate is expired.	Yes
CERT_IUT_A2_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - the certificate is not valid yet.	Yes
CERT_IUT_A3_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - application permissions: o DENM with all SSP (0x01FFFFFF); o GN-MGMT.	Yes
CERT_IUT_A4_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - application permissions: o CAM with all SPP (0x01FFFC); o GN-MGMT.	Yes
CERT_IUT_C1_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - signer digest of the CERT_IUT_CC_AA; - rectangular region restriction outside of the IUT point.	Yes
CERT_IUT_C_AT_8	Same as CERT_IUT_A_ATCERT_IUT_A_AT, excepting the following: - rectangular region restriction contains 8 elements.	Yes
CERT_TS_A_AT	Same as CERT_IUT_A_ATCERT_IUT_A_AT To be used on the Test System side.	Yes
CERT_TS_A_B_AT	Same as CERT_IUT_A_ATCERT_TS_A_AT, excepting the following: - verification key with Brainpool P256r1 curve.	Yes
CERT_TS_A_B3_AT	Same as CERT_IUT_A_ATCERT_TS_A_AT, excepting the following: - verification key with Brainpool P384r1 curve.	Yes
CERT_TS_B_AT	Same as CERT_IUT_A_ATCERT_TS_A_AT, excepting the following: - circular region restriction with a radius of 5 km from the IUT point. To be used on the Test System side.	Yes
CERT_TS_B1_AT	Same as CERT_IUT_A_ATCERT_IUT_A_B_AT, excepting the following: - circular region restriction with a radius of 5 km from the base point. To be used on the Test System side.	Yes
CERT_TS_C_AT	Same as CERT_IUT_A_ATCERT_TS_A_AT, excepting the following: - rectangular region restriction with the side of 5 km and center at the IUT point. To be used on the Test System side.	Yes
CERT_TS_D_AT	Same as CERT_IUT_A_ATCERT_TS_A_AT, excepting the following: - polygonal region restriction including the IUT position.	Yes
CERT_TS_E_AT	Same as CERT_IUT_A_ATCERT_TS_A_AT, excepting the following: - identified region restriction including the IUT point.	Yes
CERT_TS_F_AT	Same as CERT_IUT_A_ATCERT_TS_A_AT To be used on the Test System side.	No

Authorization ticket	Content	To be installed on the IUT
CERT_TS_F_AT_IMP	Same as CERT_IUT_A_AT, with imlicit certificate type To be used on the Test System side.	No
CERT_TS_F3_AT	Same as CERT_TS_F_AT, excepting the following: - verification key with Brainpool P384r1 curve. To be used on the Test System side.	No
CERT_TS_F3_AT_IMP	Same as CERT_TS_F_AT_IMP, excepting the following: - verification key with Brainpool P256r1 curve. To be used on the Test System side.	No

6.2 Sending behaviour

6.2.1 General sending behaviour

6.2.1.1 Check the message protocol version

TP Id	TP_SEC_ITSS_SND_MSG_01_BV
Summary	Check that the IUT sends a secured message containing protocol version set to 3
Reference	ETSI TS 103 097 [1], clause 5.1 IEEE Std 1609.2 [2], clause 6.3.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state ensure that when the IUT is requested to send a secured message then the IUT sends a EtsiTs103097Data containing protocolVersion indicating value '3'	

6.2.2 CAM profile

6.2.2.1 Check that secured CAM is signed

TP Id	TP_SEC_ITSS_SND_CAM_01_BV
Summary	Check that IUT sends the secured CAM using SignedData container
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing content containing signedData	

6.2.2.2 Check secured CAM AID value

TP Id	TP_SEC_ITSS_SND_CAM_02_BV
Summary	Check that IUT sends the secured CAM containing the HeaderInfo field psid set to 'AID_CAM'
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing content containing signedData containing tbsData containing headerInfo containing psid indicating 'AID_CAM'	

6.2.2.3 Check header fields

TP Id	TP_SEC_ITSS_SND_CAM_03_BV
Summary	Check that IUT sends the secured CAM with the HeaderInfo containing generationTime and does not contain expiryTime, generationLocation, encryptionKey, p2pcdLearningRequest, missingCrlIdentifier
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing content containing signedData containing tbsData containing headerInfo containing generationTime and not containing expiryTime and not containing generationLocation, and not containing encryptionKey and not containing p2pcdLearningRequest and not containing missingCrlIdentifier	

6.2.2.4 Check signer information

TP Id	TP_SEC_ITSS_SND_CAM_04_BV
Summary	Check that IUT sends the secured CAM containing signer containing either certificate or digest Check that signing certificate has permissions to sign CAM messages
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.1 IEEE Std 1609.2 [2], clause 6.3.4
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing content containing signedData containing signer containing digest or containing certificate containing id indicating 'none' containing toBeSigned containing appPermissions containing the item of type PsidSsp containing psid indicating AID_CAM and not containing certIssuePermissions	

TP Id	TP_SEC_ITSS_SND_CAM_05_BV
Summary	Check that IUT calculates the digest of certificate using proper hash algorithm Check that IUT canonicalizes certificates before hash calculation
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.1 IEEE Std 1609.2 [2], clause 6.3.4
PICS Selection	PICS_GN_SECURITY AND ***X_PICS***

Expected behaviour

with
the IUT is authorized with AT certificate (***X_CERTIFICATE***)
and the IUT is configured to send more than one CAM per second
and the IUT having sent a secured CAM
containing signer
containing certificate
indicating ***X_CERTIFICATE***
containing verifyKeyIndicator
containing verificationKey
containing ***X_KEY***
ensure that
when
the IUT is requested to send a subsequent secured CAM
containing signer
containing digest
then
the IUT sends a message of type EtsiTs103097Data
containing content
containing signedData
containing signer
containing digest
indicating last 8 bytes of the Hash value calculated using ***X_HASH*** algorithm

Permutation table

XX	***X_CERTIFICATE***	***X_KEY***	***X_HASH***	***X_PICS***
A	CERT_IUT_A_AT	ecdsaNistP256	SHA-256	
AN	CERT_IUT_A_N_AT	ecdsaNistP256 (uncompressed)	SHA-256	
B	CERT_IUT_A_B_AT	ecdsaBrainpoolP256r1	SHA-256	PICS_SEC_BRAINPOOL_P256R1
BN	CERT_IUT_A_B_N_AT	ecdsaBrainpoolP256r1 (uncompressed)	SHA-256	PICS_SEC_BRAINPOOL_P256R1
C	CERT_IUT_A_B3_AT	ecdsaBrainpoolP384r1	SHA-384	PICS_SEC_SHA384 AND PICS_SEC_BRAINPOOL_P384R1
CN	CERT_IUT_A_B3_N_AT	ecdsaBrainpoolP384r1 (uncompressed)	SHA-384	PICS_SEC_SHA384 AND PICS_SEC_BRAINPOOL_P384R1

TP Id	TP_SEC_ITSS_SND_CAM_06_BV
Summary	Check that IUT sends the secured CAM containing the signing certificate when over the time of one second no other secured CAM contained the certificate was sent
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY

Expected behaviour

with
the IUT is authorized with AT certificate (CERT_IUT_A_AT)
and the IUT is configured to send more than one CAM per second
and the IUT having sent a secured CAM
containing generationTime
indicating TIME_LAST
ensure that
when
the IUT is sending secured CAM as a message of type EtsiTs103097Data
containing signer
containing certificate
then
this message is
containing headerInfo
containing generationTime
indicating TIME (TIME >= TIME_LAST + 1 sec)

TP Id	TP_SEC_ITSS_SND_CAM_07_BV
Summary	Check that IUT sends the secured CAM containing the signing certificate when the timeout of one second has been expired after the previous CAM containing the certificate
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	

```
with
   the IUT is authorized with AT certificate (CERT_IUT_A_AT)
   and the IUT is configured to send more than one CAM per second
   and the IUT having sent a secured CAM
      containing signer
         containing certificate
      and containing generationTime
         indicating TIME_LAST
ensure that
   when
      the IUT is sending a secured CAM as a message of type EtsiTs103097Data
         containing generationTime
            indicating TIME >= TIME_LAST + 1 sec
   then
      this message is
         containing certificate
```

6.2.2.5 Check that IUT sends certificate to unknown ITS-S

TP Id	TP_SEC_ITSS_SND_CAM_08_BV
Summary	Check that IUT sends the secured CAM containing the signing certificate when the IUT received a CAM from an unknown ITS-S
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	

```
with
   the IUT is authorized with AT certificate (CERT_IUT_A_AT)
   and the IUT is configured to send more than one CAM per second
   and the IUT having already sent secured CAM
      containing certificate
      at TIME_1
   and the IUT having received a message of type EtsiTs103097Data
      containing signedData
         containing signer
            containing digest
               indicating HashedId8 value
                  referencing an unknown certificate (CERT_TS_F_AT)
                     at TIME_2 (TIME_1 < TIME_2 < TIME_1+1 sec)
ensure that
   when
      the IUT is requested to send secured CAM
            at TIME_3 (TIME_1 < TIME_2 < TIME_3 < TIME_1 + 1 sec)
   then
      the IUT sends a message of type EtsiTs103097Data
         containing signedData
            containing signer
               containing certificate
```

6.2.2.6 Check that IUT restarts the timer when the certificate has been sent

TP Id	TP_SEC_ITSS_SND_CAM_09_BV
Summary	Check that IUT restarts the certificate sending timer when the signing certificate was sent
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) and the IUT is configured to send more than one CAM per second and the IUT having already sent secured CAM containing signer containing certificate at TIME_1 and the IUT having received a secured CAM containing signer containing digest indicating HashID8 value referencing an unknown certificate (CERT_TS_F_AT) at TIME_2 (TIME_1 + 0,3 sec of tolerance) and the IUT having sent secured CAM containing signer containing certificate at TIME_3 (TIME_3 > TIME_2) ensure that when the IUT is sending the next secured CAM containing signedData containing signer containing certificate at TIME_4 then the difference between TIME_4 and TIME_3 is about 1 sec	

6.2.2.7 Check sending certificate request for unknown certificate

TP Id	TP_SEC_ITSS_SND_CAM_10_BV
Summary	Check that the IUT sends certificate request when it receives secured CAM containing digest of unknown certificate as a message signer
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.1.2
PICS Selection	PICS_GN_SECURITY, PICS_SEC_P2P_AT_DISTRIBUTION
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) and the IUT has receiving a EtsiTs103097Data containing signer containing digest indicating HashedId8 value **DIGEST_F** referencing an unknown certificate (CERT_TS_F_AT) ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing inlineP2pcdRequest containing HashedId3 value indicating last 3 octets of **DIGEST_F**	

TP Id	TP_SEC_ITSS_SND_CAM_11_BV_***XX***
Summary	Check that the IUT sends certificate request when it receives secured CAM containing certificate signed by unknown AA certificate
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.1.2
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_P2P_AA_DISTRIBUTION AND ***X_PICS***

Expected behaviour

with
the IUT is authorized with AT certificate (CERT_IUT_A_AT)
and the IUT has receiving a message of type EtsiTs103097Data
containing signer
containing certificate
containing issuer
containing ***X_FIELD_1***
indicating HashedId8 value **DIGEST_F**
referencing an unknown certificate (***X_CERT***CERT_TS_F_AT)
ensure that
when
the IUT is requested to send secured CAM
then
the IUT sends a message of type EtsiTs103097Data
containing signedData
containing tbsData
containing headerInfo
containing inlineP2pcdRequest
containing HashedId3 value
indicating last 3 octets of **DIGEST_F**

Permutation table

XX	**X_FIELD_1**	**X_CERT**	**X_PICS**
A	sha256AndDigest	CERT_TS_F_AT	
B	sha384AndDigest	CERT_TS_F3_AT	PICS_SEC_SHA384

6.2.2.8 Check that IUT sends AT certificate when requested

TP Id	TP_SEC_ITSS_SND_CAM_12_BV
Summary	Check that IUT sends the secured CAM containing the signing certificate when it received a CAM containing a request for unrecognized certificate that matches with the currently used AT certificate ID of the IUT
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.2.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_P2P_AT_DISTRIBUTION
Expected behaviour	

with
the IUT is authorized with AT certificate (CERT_IUT_A_AT)
and the IUT is configured to send more than one CAM per second
and the IUT having already sent secured CAM
containing signer
containing certificate
at TIME_1
and the IUT having received a secured CAM
containing headerInfo
containing inlineP2pcdRequest
containing HashedId3
indicating last 3 octets of currently used AT certificate (**HASHED_ID_3)**
at TIME_2 (TIME_1 < TIME_2 < TIME_1+1 sec)
ensure that
when
the IUT is requested to send a CAM
at TIME_3 (TIME_1 < TIME_2 < TIME_3 < TIME_1+1 sec)
then
the IUT sends a SecuredMessage of type EtsiTs103097Data
containing signer
and containing certificate
referenced by the HashedId3 value **HASHED_ID_3**

6.2.2.9 Check that IUT sends AA certificate when requested

TP Id	TP_SEC_ITSS_SND_CAM_13_BV
Summary	Check that IUT sends the secured CAM containing the AA certificate in the requestedCertificate headerInfo field when it received a CAM containing a request for unrecognized certificate that matches with the currently used AA certificate ID of the IUT
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.2.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_P2P_AT_DISTRIBUTION
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) issued by the AA certificate (CERT_IUT_A_AA) and the IUT is configured to send more than one CAM per second and the IUT having already sent a secured CAM containing signer containing certificate at TIME_1 and the IUT having received a secured CAM containing headerInfo containing inlineP2pcdRequest containing HashedId3 value indicating last 3 octets of the digest of CERT_IUT_A_AA at TIME_2 (TIME_1 < TIME_2 < TIME_1+1 sec) ensure that when the IUT is requested to send a secured CAM at TIME_3 (TIME_1 < TIME_2 < TIME_3 < TIME_1+1 sec) then the IUT sends a SecuredMessage of type EtsiTs103097Data containing headerInfo containing requestedCertificate indicating requested AA certificate CERT_IUT_A_AA	

TP Id	TP_SEC_ITSS_SND_CAM_14_BV
Summary	Check that IUT sends the secured CAM containing the AA certificate in the requestedCertificate headerInfo field when it received a CAM containing a request for unrecognized certificate that matches with the known AA certificate ID which is not currently used by the IUT
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.2.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_P2P_AA_DISTRIBUTION
Expected behaviour	

with
the IUT is authorized with AT certificate (CERT_IUT_A_AT)
and the IUT is configured to send more than one CAM per second
and the IUT is configured to know the AA certificate (CERT_IUT_A_N_AA)
and the IUT has already sent secured CAM
containing signer
containing certificate
at TIME_1
and the IUT having received a secured CAM
containing headerInfo
containing inlineP2pcdRequest
containing HashedId3 value
indicating last 3 octets of the digest of CERT_IUT_A_N_AA
which is not an issuer of currently used AT certificate
at TIME_2 (TIME_1 < TIME_2 < TIME_1+1 sec)
ensure that
when
the IUT is requested to send a secured CAM
at TIME_3 (TIME_1 < TIME_2 < TIME_3 < TIME_1+1 sec)
then
the IUT sends a SecuredMessage of type EtsiTs103097Data
containing headerInfo
containing requestedCertificate
indicating requested AA certificate (CERT_IUT_A_N_AA)

TP Id	TP_SEC_ITSS_SND_CAM_15_BV
Summary	Check that the IUT does not send a secured CAM containing the AA certificate in the requestedCertificate headerInfo field when it was previously requested and already received from another ITS-S
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.2.3
PICS Selection	PICS_GN_SECURITY, PICS_SEC_P2P_AA_DISTRIBUTION
Expected behaviour	

with
  the IUT is authorized with AT certificate (CERT_IUT_A_AT)
    issued by the AA certificate (CERT_IUT_A_AA)
  and the IUT is configured to send more than one CAM per second
  and the IUT having already sent secured CAM
    containing signer
      containing certificate
    at TIME_1
  and the IUT having received a secured CAM
    containing headerInfo
      containing inlineP2pcdRequest
        containing HashedId3 value
          indicating last 3 octets of the digest of CERT_IUT_A_AA
    at TIME_2 (TIME_1 < TIME_2 < TIME_1 + 0,8 sec)
  and the IUT having received a secured CAM
    containing headerInfo
      containing requestedCertificate
        indicating requested AA certificate (CERT_IUT_A_AA)
  at TIME_3 (TIME_2 < TIME_3 < TIME_2 + 0,1 sec)
ensure that
  when
    the IUT is requested to send a secured CAM
      at TIME_4 (TIME_3 < TIME_4 < TIME_1 + 0,9 sec)
  then
    the IUT sends a SecuredMessage of type EtsiTs103097Data
      containing headerInfo
        not containing requestedCertificate

TP Id	TP_SEC_ITSS_SND_CAM_16_BV
Summary	Check that the IUT does not send a secured CAM containing the AA certificate in the requestedCertificate headerInfo field when it contains certificate in the signer field
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.2.3
PICS Selection	PICS_GN_SECURITY, PICS_SEC_P2P_AA_DISTRIBUTION
Expected behaviour	

with
  the IUT is authorized with AT certificate (CERT_IUT_A_AT)
    issued by the AA certificate (CERT_IUT_A_AA)
  and the IUT is configured to send more than one CAM per second
  and the IUT having already sent a secured CAM
    containing signer
      containing certificate
    at TIME_1
  and the IUT having received a SecuredMessage
    containing headerInfo
      containing inlineP2pcdRequest
        containing HashedId3 value
          indicating last 3 octets of the digest of CERT_IUT_A_AA
  at TIME_2 (TIME2 = TIME_1 + 0,9 sec)
ensure that
  when
    the IUT is requested to send a secured CAM
      at TIME_3 (TIME_2 < TIME_3 < TIME_1 + 1 sec)
  then
    the IUT sends a SecuredMessage of type EtsiTs103097Data
      containing signer
        containing certificate
      and containing headerInfo
        not containing requestedCertificate

TP Id	TP_SEC_ITSS_SND_CAM_17_BV
Summary	Check that the IUT sends a secured CAM containing the AA certificate in the requestedCertificate headerInfo field with the next CAM containing digest as a signer info
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.9 and 8.2.4.2.3
PICS Selection	PICS_GN_SECURITY, PICS_SEC_P2P_AA_DISTRIBUTION
Expected behaviour	

with
  the IUT is authorized with AT certificate (CERT_IUT_A_AT)
    issued by the AA certificate (CERT_IUT_A_AA)
  and the IUT is configured to send more than one CAM per second
  and the IUT having already sent secured CAM
    containing signer
      containing certificate
    at TIME_1
  and the IUT having received a SecuredMessage of type EtsiTs103097Data
    containing headerInfo
      containing inlineP2pcdRequest
        containing HashedId3 value
          indicating last 3 octets of the digest of CERT_IUT_A_AA
    at TIME_2 (TIME_1+0,9 sec < TIME2 < TIME_1 + 1 sec)
ensure that
  when
    the IUT is sending a first subsequent secured CAM
      containing signer
        containing digest
  then
    this message
      containing headerInfo
        containing requestedCertificate
          indicating requested AA certificate CERT_IUT_A_AA

6.2.2.10 Check generation time

TP Id	TP_SEC_ITSS_SND_CAM_18_BV
Summary	Check that IUT sends the secured CAM containing generation time and this time is inside the validity period of the signing certificate Check that message generation time value is realistic
Reference	ETSI TS 103 097 [1], clause 7.1.1 IEEE Std 1609.2 [2], clauses 5.2.3.2.2, 5.2.4.2.2 and 5.2.4.2.3
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send CAM containing certificate then the IUT sends a SecuredMessage of type EtsiTs103097Data containing headerInfo containing generationTime indicating GEN_TIME (CUR_TIME - 5 min <= GEN_TIME <= CUR_TIME + 5 min) and containing signer containing certificate containing toBeSigned containing validityPeriod containing start indicating value X_START_VALIDITY (X_START_VALIDITY <= GEN_TIME) and containing duration indicating value > GEN_TIME - X_START_VALIDITY	

6.2.2.11 Check payload

TP Id	TP_SEC_ITSS_SND_CAM_19_BV
Summary	Check that IUT sends the secured CAM containing the 'data' field in signed data payload, containing the EtsiTs103097Data of type unsecured, contained the CAM payload
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data contains content contains signedData containing tbsData containing payload containing data containing content containing unsecuredData containing not-empty data	

6.2.2.12 Check signing permissions

TP Id	TP_SEC_ITSS_SND_CAM_20_BV
Summary	Check that the IUT sends the secured CAM signed with the certificate containing appPermissions allowing to sign CA messages
Reference	ETSI TS 103 097 [1], clause 7.2.1 IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing signer containing certificate containing appPermissions containing an item of type PsidSsp containing psid = AID_CAM	

6.2.2.13 Check signature

TP Id	TP_SEC_ITSS_SND_CAM_21_BV_***XX***
Summary	Check that IUT sends the secured CAM containing signature Check that the signature is calculated over the right fields and using right hash algorithm by cryptographically verifying the signature
Reference	ETSI TS 103 097 [1], clauses 5.2, 7.1.1 IEEE Std 1609.2 [2], clauses 5.3.1, 6.3.4, 6.3.29, 6.3.30 and 6.3.31
PICS Selection	PICS_GN_SECURITY AND ***X_PICS***
Expected behaviour	
with the IUT is authorized with AT certificate (***X_CERTIFICATE***) containing verifyKeyIndicator containing verificationKey containing ***X_KEY*** indicating KEY ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing signedData containing signer containing digest referencing the certificate ***X_CERTIFICATE*** or containing certificate indicating ***X_CERTIFICATE*** and containing signature containing ***X_SIGNATURE*** verifiable using KEY	

Permutation table

XX	***X_CERTIFICATE***	***X_KEY***	***X_SIGNATURE***	***X_PICS***
A	CERT_IUT_A_AT	ecdsaNistP256	ecdsaNistP256Signature	
B	CERT_IUT_A_B_AT	ecdsaBrainpoolP256r1	ecdsaBrainpoolP256r1Signature	PICS_SEC_BRAINPOOL_P256R1
C	CERT_IUT_A_B3_AT	ecdsaBrainpoolP384r1	ecdsaBrainpoolP384r1Signature	PICS_SEC_SHA384 AND PICS_SEC_BRAINPOOL_P384R1

TP Id	TP_SEC_ITSS_SND_CAM_22_BV_**XX**
Summary	Check that IUT sends the secured CAM containing signature containing the ECC point of type set to either compressed_lsb_y_0, compressed_lsb_y_1 or x_coordinate_only
Reference	ETSI TS 103 097 [1], clauses 5.2, 7.1.1 IEEE Std 1609.2 [2], clauses 6.3.30 and 6.3.31
PICS Selection	PICS_GN_SECURITY AND ***X_PICS***

Expected behaviour

with
  the IUT is authorized with AT certificate (***X_CERTIFICATE***)
ensure that
  when
    the IUT is requested to send a secured CAM
  then
    the IUT sends a message of type EtsiTs103097Data
      containing signedData
        containing signature
          containing ***X_SIGNATURE***
            containing rSig
              containing x-only
              or containing compressed-y-0
              or containing compressed-y-1

Permutation table

XX	***X_CERTIFICATE***	***X_SIGNATURE***	***X_PICS***
A	CERT_IUT_A_AT	ecdsaNistP256Signature	
B	CERT_IUT_A_B_AT	ecdsaBrainpoolP256r1Signature	PICS_SEC_BRAINPOOL_P256R1
C	CERT_IUT_A_B3_AT	ecdsaBrainpoolP384r1Signature	PICS_SEC_SHA384 AND PICS_SEC_BRAINPOOL_P384R1

6.2.2.14 Check support for certificate content

TP Id	TP_SEC_ITSS_SND_CAM_23_BV
Summary	Check that IUT supports at least 8 items in the appPermissions component of the certificate
Reference	IEEE Std 1609.2 [2], clause 6.4.8
PICS Selection	PICS_GN_SECURITY

Expected behaviour

with
  the IUT is authorized with AT certificate (CERT_IUT_A_AT_A8)
    containing toBeSigned
      containing appPermissions
        containing 8 entries
          indicating the last item
            containing psid
              indicating the 'AID_CAM'
ensure that
  when
    the IUT is requested to send a secured CAM
  then
    the IUT sends a message of type EtsiTs103097Data
      containing content
        containing signedData
          containing tbsData
            containing headerInfo
              containing psid
                indicating 'AID_CAM'

TP Id	TP_SEC_ITSS_SND_CAM_24_BV
Summary	Check that IUT supports at least 8 items in the certIssuePermissions component of the certificate
Reference	IEEE Std 1609.2 [2], clause 6.4.8
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT_A8) containing appPermissions conformed to the certIssuePermissions issued by AA certificate (CERT_IUT_A_AA_A8) containing toBeSigned containing certIssuePermissions containing 8 entries indicating the last item containing psid indicating the 'AID_CAM' ensure that when the IUT is requested to send a secured CAM then the IUT sends a message of type EtsiTs103097Data containing content containing signedData containing tbsData containing headerInfo containing psid indicating 'AID_CAM'	

6.2.2.15 Check certificate consistency conditions

TP Id	TP_SEC_ITSS_SND_CAM_23_BV
Summary	Check that IUT does not send secured CAMs if IUT is authorized with AT certificate does not allow sending messages in this location
Reference	IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_C1_AT) containing region indicating rectangular region not containing current IUT position and the IUT has no other installed AT certificates ensure that when the IUT is requested to send a secured CAM then the IUT does not send CAM	

TP Id	TP_SEC_ITSS_SND_CAM_24_BV
Summary	Check that IUT does not send the secured CAM if IUT is configured to use an AT certificate without region validity restriction and generation location is outside of the region of the issuing AA certificate
Reference	IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_CA3_AT) not containing region and issued by the AA certificate (CERT_IUT_C3_AA) containing region indicating rectangular region not containing current IUT position ensure that when the IUT is requested to send a secured CAM then the IUT does not send CAM	

TP Id	TP_SEC_ITSS_SND_CAM_25_BV
Summary	Check that IUT does not send secured CAMs if all AT certificates installed on the IUT was expired
Reference	IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A1_AT) containing validityPeriod indicating start + duration < CURRENT_TIME and the IUT has no other installed AT certificates ensure that when the IUT is requested to send a secured CAM then the IUT does not send CAM	

TP Id	TP_SEC_ITSS_SND_CAM_26_BV
Summary	Check that IUT does not send secured CAMs if all AT certificates installed on the IUT have the starting time in the future
Reference	IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A2_AT) containing validityPeriod indicating start > CURRENT_TIME and the IUT has no other installed AT certificates ensure that when the IUT is requested to send a secured CAM then the IUT does not send CAM	

TP Id	TP_SEC_ITSS_SND_CAM_27_BV
Summary	Check that IUT does not send secured CAMs if IUT does not possess an AT certificate allowing sending CAM by its appPermissions
Reference	IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A3_AT) containing appPermissions not containing PsidSSP containing psid indicating AID_CAM and the IUT has no other installed AT certificates ensure that when the IUT is requested to send a secured CAM then the IUT does not send CAM	

6.2.3 DENM profile

6.2.3.1 Check secured DENM is signed

TP Id	TP_SEC_ITSS_SND_DENM_01_BV
Summary	Check that IUT sends the secured DENM using SignedData container
Reference	ETSI TS 103 097 [1], clause 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured DENM then the IUT sends a EtsiTs103097Data containing content containing signedData	

6.2.3.2 Check secured DENM AID value

TP Id	TP_SEC_ITSS_SND_DENM_02_BV
Summary	Check that IUT sends the secured DENM containing the HeaderInfo field psid set to 'AID_DENM'
Reference	ETSI TS 103 097 [1], clause 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured DENM then the IUT sends a EtsiTs103097Data containing content containing signedData containing tbsData containing headerInfo containing psid indicating 'AID_DENM'	

6.2.3.3 Check header fields

TP Id	TP_SEC_ITSS_SND_DENM_03_BV
Summary	Check that IUT sends the secured DENM with the HeaderInfo containing generationTime and generationLocation and does not contain expiryTime, encryptionKey, p2pcdLearningRequest, missingCrlIdentifier, inlineP2pcdRequest, requestedCertificate
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured DENM then the IUT sends a EtsiTs103097Data containing content containing signedData containing tbsData containing headerInfo containing generationTime and containing generationLocation, and not containing expiryTime and not containing encryptionKey and not containing p2pcdLearningRequest and not containing missingCrlIdentifier and not containing inlineP2pcdRequest and not containing requestedCertificate	

6.2.3.4 Check signer information

TP Id	TP_SEC_ITSS_SND_DENM_04_BV
Summary	Check that IUT sends the secured DENM containing signer containing certificate
Reference	ETSI TS 103 097 [1], clause 7.1.2 IEEE Std 1609.2 [2], clause 6.3.4
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured DENM then the IUT sends a EtsiTs103097Data containing content containing signedData containing signer containing certificate containing toBeSigned containing appPermissions containing the item of type PsidSsp containing psid indicating AID_DENM	

6.2.3.5 Check generation time

TP Id	TP_SEC_ITSS_SND_DENM_05_BV
Summary	Check that IUT sends the secured DENM containing generation time and this time is inside the validity period of the signing certificate Check that message generation time value is realistic
Reference	ETSI TS 103 097 [1], clause 7.1.2 IEEE Std 1609.2 [2], clauses 5.2.3.2.2, 5.2.4.2.2 and 5.2.4.2.3
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured DENM then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing generationTime indicating GEN_TIME (CUR_TIME - 10min <= GEN_TIME <= CUR_TIME + 10 min) and containing signer containing certificate containing toBeSigned containing validityPeriod containing start indicating value X_START_VALIDITY (X_START_VALIDITY <= GEN_TIME) and containing duration indicating value > GEN_TIME - X_START_VALIDITY	

6.2.3.6 Check generation location

TP Id	TP_SEC_ITSS_SND_DENM_06_BV
Summary	Check that IUT sends the secured DENM containing generation location when signing certificate chain does not have any region restriction
Reference	ETSI TS 103 097 [1], clause 7.1.2 IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_CERTIFICATE_SELECTION
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_A_AT) containing toBeSigned not containing region and issued by the certificate AA (CERT_IUT_A_AA) containing toBeSigned not containing region and issued by the certificate RCA (CERT_IUT_A_RCA) containing toBeSigned not containing region ensure that when the IUT is requested to send a secured DENM then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing generationLocation	

TP Id	TP_SEC_ITSS_SND_DENM_07_BV_***XX***
Summary	Check that IUT sends the secured DENM containing generation location which is inside the region defined by the validity restriction of the certificate pointed by the message signer
Reference	ETSI TS 103 097 [1], clause 7.1.2 IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_CERTIFICATE_SELECTION AND ***X_PICS***

Expected behaviour

with
the IUT has been authorized with the AT certificate (***X_AT_CERTIFICATE***)
containing toBeSigned
containing region
containing ***X_FIELD***
indicating REGION
ensure that
when
the IUT is requested to send a secured DENM
then
the IUT sends a message of type EtsiTs103097Data
containing headerInfo
containing generationLocation
indicating value inside the REGION

Permutation Table

_XX	***X_FIELD***	***X_AT_CERTIFICATE***	***X_PICS***
B	circularRegion	CERT_IUT_B_AT	PICS_SEC_CIRCULAR_REGION
C	rectangularRegion	CERT_IUT_C_AT	PICS_SEC_RECTANGULAR_REGION
D	polygonalRegion	CERT_IUT_D_AT	PICS_SEC_POLYGONAL_REGION
E	identifiedRegion	CERT_IUT_E_AT	PICS_SEC_IDENTIFIED_REGION

TP Id	TP_SEC_ITSS_SND_DENM_09_BV
Summary	Check that IUT sends the secured DENM containing generation location which is inside the identified region defined by the validity restriction of the AA certificate used to sign the certificate pointed by the message signer does not contain any region restriction
Reference	ETSI TS 103 097 [1], clause 7.1.2 IEEE Std 1609.2 [2], clauses 5.2.3.2.2 and 6.4.8
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_CERTIFICATE_SELECTION

Expected behaviour

with
the IUT has been authorized with the AT certificate (CERT_IUT_CA1_AT)
containing toBeSigned
not containing region
and issued by the certificate AA (CERT_IUT_CC_AA)
containing toBeSigned
containing circularRegion
indicating REGION
and issued by the certificate RCA (CERT_IUT_C_RCA)
containing toBeSigned
containing circularRegion
indicating REGION
ensure that
when
the IUT is requested to send a secured DENM
then
the IUT sends a message of type EtsiTs103097Data
containing headerInfo
containing generationLocation
indicating value inside the REGION

TP Id	TP_SEC_ITSS_SND_DENM_10_BV
Summary	Check that IUT sends the secured DENM containing generation location which is inside the identified region defined by the validity restriction of the root certificate when subordinate AA and AT certificates do not contain any region restriction
Reference	ETSI TS 103 097 [1], clause 7.1.2 IEEE Std 1609.2 [2], clauses 5.2.3.2.2 and 6.4.8
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_CERTIFICATE_SELECTION
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_CA2_AT) containing toBeSigned not containing region and issued by the certificate AA (CERT_IUT_CA_AA) containing toBeSigned not containing region and issued by the certificate RCA (CERT_IUT_C_RCA) containing toBeSigned containing circularRegion indicating REGION ensure that when the IUT is requested to send a secured DENM then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing generationLocation indicating value inside the REGION	

6.2.3.7 Check payload

TP Id	TP_SEC_ITSS_SND_DENM_11_BV
Summary	Check that IUT sends the secured DENM containing the 'data' field in signed data payload, containing the EtsiTs103097Data of type unsecured, contained the DENM payload
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured DENM then the IUT sends a message of type EtsiTs103097Data contains content contains signedData containing tbsData containing payload containing data containing content containing unsecuredData containing not-empty data	

6.2.3.8 Check signing permissions

TP Id	TP_SEC_ITSS_SND_DENM_12_BV
Summary	Check that the IUT sends the secured DENM signed with the certificate containing appPermissions allowing to sign DEN messages
Reference	ETSI TS 103 097 [1], clause 7.1.2 IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY

Expected behaviour

with
the IUT has been authorized with the AT certificate (CERT_IUT_A_AT)
ensure that
when
the IUT is requested to send a secured DENM
then
the IUT sends a message of type EtsiTs103097Data
containing signer
containing certificate
containing appPermissions
containing an item of type PsidSsp
containing psid
indicating AID_DENM

6.2.3.9 Check signature

TP Id	TP_SEC_ITSS_SND_DENM_13_BV
Summary	Check that IUT sends the secured DENM containing signature Check that the signature is calculated over the right fields and using right hash algorithm by cryptographically verifying the signature
Reference	ETSI TS 103 097 [1], clauses 5.2, 7.1.2 IEEE Std 1609.2 [2], clauses 5.3.1, 6.3.4, 6.3.29, 6.3.30 and 6.3.31
PICS Selection	PICS_GN_SECURITY AND ***X_PICS***

Expected behaviour

with
the IUT is authorized with AT certificate (***X_CERTIFICATE***)
containing verifyKeyIndicator
containing verificationKey
containing ***X_KEY***
indicating KEY
ensure that
when
the IUT is requested to send a secured DENM
then
the IUT sends a message of type EtsiTs103097Data
containing signedData
containing signer
containing certificate
indicating ***X_CERTIFICATE***
containing verifyKeyIndicator
containing verificationKey
containing ***X_KEY***
indicating KEY
and containing signature
containing ***X_SIGNATURE***
verifiable using KEY

Permutation table

XX	*X_CERTIFICATE*	*X_KEY*	*X_SIGNATURE*	*X_PICS*
A	CERT_IUT_A_AT	ecdsaNistP256	ecdsaNistP256Signature	
B	CERT_IUT_A_B_AT	ecdsaBrainpoolP256r1	ecdsaBrainpoolP256r1Signature	PICS_SEC_BRAINPOOL_P256R1
C	CERT_IUT_A_B3_AT	ecdsaBrainpoolP384r1	ecdsaBrainpoolP384r1Signature	PICS_SEC_SHA384 AND PICS_SEC_BRAINPOOL_P384R1

6.2.3.10 Check support for certificate content

TP Id	TP_SEC_ITSS_SND_DENM_14_BV
Summary	Check that the IUT supports at least 8 entries in the rectangular certificate validity region in the AT certificate
Reference	IEEE Std 1609.2 [2], clause 6.4.17
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_RECTANGULAR_REGION
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_C_AT_8) containing toBeSigned containing region containing rectangularRegion containing 8 entries containing an entry (***ENTRY***) containing current IUT position ensure that when the IUT is requested to send a secured DENM then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing generationLocation indicating position inside the ***ENTRY***	

TP Id	TP_SEC_ITSS_SND_DENM_15_BV
Summary	Check that the IUT supports at least 8 points in the polygonal certificate validity region in the AT certificate
Reference	IEEE Std 1609.2 [2], clause 6.4.17
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_POLYGONAL_REGION
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_D_AT_8) containing toBeSigned containing region containing polygonalRegion containing 8 entries indicating polygon ***P*** and the IUT's position is inside the polygon ***P*** ensure that when the IUT is requested to send a secured DENM then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing generationLocation indicating position inside the ***P***	

TP Id	TP_SEC_ITSS_SND_DENM_16_BV
Summary	Check that the IUT supports at least 8 points in the polygonal certificate validity region in the AT certificate
Reference	IEEE Std 1609.2 [2], clause 6.4.17
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_IDENTIFIED_REGION
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_E_AT_8) containing toBeSigned containing region containing identifiedRegion containing 8 entries containing one of the items (***I***) containing current IUT position ensure that when the IUT is requested to send a secured DENM then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing generationLocation indicating position inside the ***I***	

6.2.3.11 Check certificate consistency conditions

TP Id	TP_SEC_ITSS_SND_DENM_17_BV
Summary	Check that IUT does not send secured DENMs if IUT does not possess an AT certificate allowing sending messages in this location
Reference	IEEE Std 1609.2 [2], clause 6.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT has been authorized with the AT certificate CERT_IUT_C1_AT) containing region indicating rectangular region not containing current IUT position ensure that when the IUT is requested to send a secured DENM then the IUT does not send DENM	

TP Id	TP_SEC_ITSS_SND_DENM_18_BV
Summary	Check that IUT does not send the secured DENM if IUT is configured to use an AT certificate without region validity restriction and generation location is outside of the region of the issuing AA certificate
Reference	IEEE Std 1609.2 [2], clause 6.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_CA3_AT) not containing region and issued by the AA certificate (CERT_IUT_C3_AA) containing region indicating rectangular region not containing current IUT position ensure that when the IUT is requested to send a secured DENM then the IUT does not send DENM	

TP Id	TP_SEC_ITSS_SND_DENM_19_BV
Summary	Check that IUT does not send secured DENMs if all AT certificates installed on the IUT are expired
Reference	IEEE Std 1609.2 [2], clause 6.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A1_AT) containing validityPeriod indicating start + duration < CURRENT_TIME and the IUT has no other installed AT certificates ensure that when the IUT is requested to send a secured DENM then the IUT does not send DENM	

TP Id	TP_SEC_ITSS_SND_DENM_20_BV
Summary	Check that IUT does not send secured DENMs if all AT certificates installed on the IUT have the starting time in the future
Reference	IEEE Std 1609.2 [2], clause 6.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_A2_AT) containing validityPeriod indicating start > CURRENT_TIME and IUT has no other certificates installed ensure that when the IUT is requested to send a secured DENM then the IUT does not send DENM	

TP Id	TP_SEC_ITSS_SND_DENM_21_BV
Summary	Check that IUT does not send secured DENMs if IUT does not possess an AT certificate allowing sending DENM by its appPermissions
Reference	IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_A4_AT) containing appPermissions not containing PsidSSP containing psid indicating AID_DENM and IUT has no other certificates installed ensure that when the IUT is requested to send a secured DENM then the IUT does not send DENM	

6.2.4 Generic signed message profile

6.2.4.1 Check that secured message is signed

TP Id	TP_SEC_ITSS_SND_GENMSG_01_BV
Summary	Check that IUT sends the secured message using signedData container
Reference	ETSI TS 103 097 [1], clause 7.1.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured Beacon then the IUT sends a message of type EtsiTs103097Data containing content containing signedData	

6.2.4.2 Check secured AID value

TP Id	TP_SEC_ITSS_SND_GENMSG_02_BV
Summary	Check that the sent Secured Message contains HeaderField its_aid that is set to other value then AID_CAM and AID_DENM
Reference	ETSI TS 103 097 [1], clause 7.1.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER
Expected behaviour	
with the IUT is authorized with AT certificate CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured Beacon then the IUT sends a message of type EtsiTs103097Data containing content containing signedData containing tbsData containing headerInfo containing psid indicating AID_GNMGMT	

6.2.4.3 Check header field

TP Id	TP_SEC_ITSS_SND_GENMSG_03_BV
Summary	Check that IUT sends the secured GeoNetworking message with the headerInfo containing generationTime
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured Beacon then the IUT sends a message of type EtsiTs103097Data containing content containing signedData containing tbsData containing headerInfo containing generationTime and not containing p2pcdLearningRequest and not containing missingCrlIdentifier	

6.2.4.4 Check that signer info is a certificate or digest

TP Id	TP_SEC_ITSS_SND_GENMSG_04_BV
Summary	Check that IUT sends the secured GeoNetworking message containing certificate or digest as a signer
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.3 IEEE Std 1609.2 [2], clause 6.3.4
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured Beacon then the IUT sends a message of type EtsiTs103097Data containing content containing signedData containing signer containing digest or containing certificate containing toBeSigned containing appPermissions containing the item of type PsidSsp containing psid indicating AID_GNMGMT	

6.2.4.5 Check generation time

TP Id	TP_SEC_ITSS_SND_GENMSG_05_BV
Summary	Check that IUT sends the secured GeoNetworking message containing generation time and this time is inside the validity period of the signing certificate Check that message generation time value is realistic
Reference	ETSI TS 103 097 [1], clauses 5.4 and 7.1.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured Beacon containing certificate then the IUT sends a message of type EtsiTs103097Data containing headerInfo containing generationTime indicating GEN_TIME (CUR_TIME - 10 min <= GEN_TIME <= CUR_TIME + 10 min) and containing signer containing certificate containing toBeSigned containing validityPeriod containing start indicating value X_START_VALIDITY (X_START_VALIDITY <= GEN_TIME) and containing duration indicating value > GEN_TIME - X_START_VALIDITY	

6.2.4.6 Check payload

TP Id	TP_SEC_ITSS_SND_GENMSG_06_BV
Summary	Check that IUT sends the secured message using the 'data' field in signed data payload, containing the EtsiTs103097Data of type unsecured, containing the data payload or using the extDataHash field containing the SHA256 hash of data payload
Reference	ETSI TS 103 097 [1], clause 7.1.3
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER
Expected behaviour	
with the IUT is authorized with AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send a secured Beacon then the IUT sends a message of type EtsiTs103097Data contains content contains signedData containing tbsData containing payload containing data containing content containing unsecuredData containing not-empty data	

6.2.4.7 Check signing permissions

TP Id	TP_SEC_ITSS_SND_GENMSG_07_BV
Summary	Check that the IUT sends the secured messages signed with the certificate containing appPermissions allowing to sign these messages
Reference	ETSI TS 103 097 [1], clause 7.1.3 IEEE Std 1609.2 [2], clause 5.2.3.2.2
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER
Expected behaviour	
with the IUT has been authorized with the AT certificate (CERT_IUT_A_AT) ensure that when the IUT is requested to send Beacon then the IUT sends a message of type EtsiTs103097Data containing signer containing certificate containing appPermissions containing an item of type PsidSsp containing psid indicating value AID_GNMGMT	

6.2.4.8 Check signature

TP Id	TP_SEC_ITSS_SND_GENMSG_08_BV
Summary	Check that IUT sends the secured GeoNetworking message containing signature Check that the signature is calculated over the right fields and using right hash algorithm by cryptographically verifying the signature
Reference	ETSI TS 103 097 [1], clauses 5.2 and 7.1.3 IEEE Std 1609.2 [2], clauses 5.3.1, 6.3.4, 6.3.29, 6.3.30 and 6.3.31
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_ITS_AID_OTHER AND ***X_PICS***
Expected behaviour	
with the IUT is authorized with AT certificate (***X_CERTIFICATE***) containing verifyKeyIndicator containing verificationKey containing ***X_KEY*** indicating KEY ensure that when the IUT is requested to send a secured Beacon then the IUT sends a message of type EtsiTs103097Data containing signedData containing signer containing digest referencing the certificate ***X_CERTIFICATE*** or containing certificate indicating ***X_CERTIFICATE*** and containing signature containing ***X_SIGNATURE*** verifiable using KEY	

Permutation table

XX	*X_CERTIFICATE*	*X_KEY*	*X_SIGNATURE*	*X_PICS*
A	CERT_IUT_A_AT	ecdsaNistP256	ecdsaNistP256Signature	
B	CERT_IUT_A_B_AT	ecdsaBrainpoolP256r1	ecdsaBrainpoolP256r1Signature	PICS_SEC_BRAINPOOL_P256R1
C	CERT_IUT_A_B3_AT	ecdsaBrainpoolP384r1	ecdsaBrainpoolP384r1Signature	PICS_SEC_SHA384 AND PICS_SEC_BRAINPOOL_P384R1

6.3 Receiving behaviour

6.3.1 Check the message protocol version

TP Id	TP_SEC_ITSS_RCV_MSG_01_BV
Summary	Check that IUT accepts a secured message containing protocol version set to a value 3
Reference	ETSI TS 103 097 [1], clause 5.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is being authorized with the certificate CERT_IUT_A_AT and the IUT current time is inside the time validity period of CERT_TS_A_AT and CERT_IUT_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data signed using CERT_TS_A_AT and containing protocolVersion indicating 3 then the IUT forwards the SecuredMessage to the Facility layers	

TP Id	TP_SEC_ITSS_RCV_MSG_01_BO
Summary	Check that IUT discards a secured message containing protocol version set to a value less than 3
Reference	ETSI TS 103 097 [1], clause 5.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is being authorized with the certificate CERT_IUT_A_AT and the IUT current time is inside the time validity period of CERT_TS_A_AT and CERT_IUT_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data signed using CERT_TS_A_AT and containing protocolVersion indicating 2 then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_MSG_02_BO
Summary	Check that IUT discards a secured message containing protocol version set to a value greater than 3
Reference	ETSI TS 103 097 [1], clause 5.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT is being authorized with the certificate CERT_IUT_A_AT and the IUT current time is inside the time validity period of CERT_TS_A_AT and CERT_IUT_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data signed using CERT_TS_A_AT and containing protocolVersion indicating 4 then the IUT discards the SecuredMessage	

6.3.2 CAM profile

6.3.2.1 Check the valid message receiving

TP Id	TP_SEC_ITSS_RCV_CAM_01_BV
Summary	Check that IUT accepts a valid secured CAM message signed with certificate
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	

```
with
    the IUT being in the 'authorized' state
    and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_AT
ensure that
    when
        the IUT is receiving a message of type EtsiTs103097Data (MSG)
            containing protocolVersion
                indicating 3
            and containing content.signedData
                containing hashId
                    indicating hash algorithm of the verification key of CERT_TS_A_AT
                and containing tbsData
                    containing payload
                        containing data
                            containing protocolVersion
                                indicating 3
                            and containing content.unsecuredData
                                containing CAM payload
                    and containing headerInfo
                        containing psid
                            indicating CAM AID value
                        and containing generationTime
                            indicating time within 2sec around the CUR_TIME
                        and NOT containing other headers
                and containing signer
                    containing certificate
                        containing 1 item of type EtsiTs103097Certificate
                            indicating CERT_TS_A_AT
                and containing signature
                    containing ecdsaNistP256Signature
                        containing rSig.x-only
                    calculated over the MSG.content.signedData.tbsData
                        using verification key of CERT_TS_A_AT
    then
        the IUT accepts the SecuredMessage
```

TP Id	TP_SEC_ITSS_RCV_CAM_02_BV
Summary	Check that IUT accepts a valid secured CAM message signed with digest
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	

```
with
    the IUT being in the 'authorized' state
    and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_AT
    and the IUT has already received the message signed with CERT_TS_A_AT
ensure that
    when
        the IUT is receiving a message of type EtsiTs103097Data
            indicating the message described in TP_SEC_ITSS_RCV_CAM_01_BV
            and containing content.signedData.signer
                containing digest
                    indicating HashedId8 value
                        referencing the CERT_TS_A_AT
    then
        the IUT accepts the SecuredMessage
```

TP Id	TP_SEC_ITSS_RCV_CAM_03_BV
Summary	Check that IUT accepts a valid secured CAM message signed with compressed signature
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data (MSG) indicating the message described in TP_SEC_ITSS_RCV_CAM_01_BV and containing content.signedData.signature containing ecdsaNistP256Signature containing rSig.compressed-y-0 or containing rSig.compressed-y-1 calculated over the MSG.content.signedData.tbsData using verification key of CERT_TS_A_AT then the IUT accepts the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_CAM_04_BV_***XX***
Summary	Check that IUT accepts a valid secured CAM message signed with certificate containing region restriction
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY AND ***X_PICS***
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time (CUR_TIME) is inside the time validity period of ***X_AT_CERTIFICATE*** and the IUT current position is inside the region restriction of ***X_AT_CERTIFICATE*** ensure that when the IUT is receiving a message of type EtsiTs103097Data (MSG) indicating the message described in TP_SEC_ITSS_RCV_CAM_01_BV and containing content.signedData containing signer containing certificate containing 1 item of type EtsiTs103097Certificate indicating ***X_AT_CERTIFICATE*** containing toBeSigned.region containing ***X_FIELD*** and containing signature containing ecdsaNistP256Signature containing rSig.x-only calculated over the MSG.content.signedData.tbsData using verification key of ***X_AT_CERTIFICATE*** then the IUT accepts the SecuredMessage	

Permutation Table

_XX	***X_FIELD***	***X_AT_CERTIFICATE***	***X_PICS***
01	circularRegion	CERT_TS_B_AT	PICS_SEC_CIRCULAR_REGION
02	rectangularRegion	CERT_TS_C_AT	PICS_SEC_RECTANGULAR_REGION
03	polygonalRegion	CERT_TS_D_AT	PICS_SEC_POLYGONAL_REGION
04	identifiedRegion	CERT_TS_E_AT	PICS_SEC_IDENTIFIED_REGION

TP Id	TP_SEC_ITSS_RCV_CAM_05_BV
Summary	Check that IUT accepts a valid secured CAM message signed using the brainpoolP256r1 algorithm
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_BRAINPOOL_P256R1
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_B_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data indicating the message described in TP_SEC_ITSS_RCV_CAM_01_BV and containing content.signedData containing signer containing certificate containing 1 item of type EtsiTs103097Certificate indicating CERT_TS_A_B_AT containing toBeSigned.verifyKeyIndicator.verificationKey containing ecdsaBrainpoolP256r1 and containing signature containing ecdsaBrainpoolP256r1Signature containing rSig.x-only calculated over the MSG.content.signedData.tbsData using verification key of CERT_TS_A_B_AT then the IUT accepts the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_CAM_06_BV
Summary	Check that IUT accepts a valid secured CAM message signed using the brainpoolP384r1 algorithm
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_BRAINPOOL_P384R1
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_B3_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data indicating the message described in TP_SEC_ITSS_RCV_CAM_01_BV and containing content.signedData containing signer containing certificate containing 1 item of type EtsiTs103097Certificate indicating CERT_TS_A_B3_AT containing toBeSigned. verifyKeyIndicator.verificationKey containing ecdsaBrainpoolP384r1 and containing signature containing ecdsaBrainpoolP384r1Signature containing rSig.x-only calculated over the MSG.content.signedData.tbsData using verification key of CERT_TS_A_B3_AT then the IUT accepts the SecuredMessage	

6.3.2.2 Check invalid HeaderInfo elements

TP Id	TP_SEC_ITSS_RCV_CAM_01_BO
Summary	Check that IUT discards a secured CAM if the HeaderInfo contains the header field an invalid Psid value
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data indicating the message described in TP_SEC_ITSS_RCV_CAM_02_BV and containing SignedData containing ToBeSignedData containing HeaderInfo containing psid **not** indicating CAM AID value then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_CAM_02_BO
Summary	Check that IUT discards a secured CAM if the HeaderInfo contains the header field generationLocation
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing psid indicating CAM AID value and containing generationLocation then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_CAM_03_BO
Summary	Check that IUT discards a secured CAM if the HeaderInfo contains the header field expiryTime
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	

With
the IUT being in the 'authorized' state
and the IUT current time is inside the time validity period of CERT_TS_A_AT
ensure that
when
the IUT is receiving a message of type EtsiTs103097Data
containing SignedData
containing ToBeSignedData
containing HeaderInfo
containing psid
indicating CAM AID value
and containing expiryTime
then
the IUT discards the SecuredMessage

TP Id	TP_SEC_ITSS_RCV_CAM_04_BO
Summary	Check that IUT discards a secured CAM if the HeaderInfo contains the header field p2pcdLearningRequest
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	

With
the IUT being in the 'authorized' state
and the IUT current time is inside the time validity period of CERT_TS_A_AT
ensure that
when
the IUT is receiving a message of type EtsiTs103097Data
containing SignedData
containing ToBeSignedData
containing HeaderInfo
containing psid
indicating CAM AID value
and containing p2pcdLearningRequest
then
the IUT discards the SecuredMessage

TP Id	TP_SEC_ITSS_RCV_CAM_05_BO
Summary	Check that IUT discards a secured CAM if the HeaderInfo contains the header field missingCrlIdentifier
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	

With
the IUT being in the 'authorized' state
and the IUT current time is inside the time validity period of CERT_TS_A_AT
ensure that
when
the IUT is receiving a message of type EtsiTs103097Data
containing SignedData
containing ToBeSignedData
containing HeaderInfo
containing psid
indicating CAM AID value
and containing missingCrlIdentifier
then
the IUT discards the SecuredMessage

TP Id	TP_SEC_ITSS_RCV_CAM_06_BO
Summary	Check that IUT discards a secured CAM if the HeaderInfo contains the header field encryptionKey
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing psid indicating CAM AID value and containing encryptionKey then the IUT discards the SecuredMessage	

6.3.2.3 Check invalid Signature elements

TP Id	TP_SEC_ITSS_RCV_CAM_07_BO
Summary	Check that IUT discards a secured CAM if the 'SignedData' contains an invalid signature algorithm
Reference	ETSI TS 103 097 [1], clause 6
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing Signature indicating invalid signature algorithm then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_CAM_08_BO
Summary	Check that IUT discards a secured CAM if the 'SignerIdentifier' contains an invalid choice
Reference	ETSI TS 103 097 [1], clause 6
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing SignerIdentifier indicating 'self' then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_CAM_09_BO
Summary	Check that IUT discards a secured CAM if the Signature cannot be verified
Reference	ETSI TS 103 097 [1], clause 6
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing Signature indicating an altered value different then the one calculated using signature calculated rules then the IUT discards the SecuredMessage	

6.3.3 DENM profile

6.3.3.1 Check the valid message receiving

TP Id	TP_SEC_ITSS_RCV_DENM_01_BV
Summary	Check that IUT accepts a valid secured DENM message signed with certificate
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data (MSG) containing protocolVersion indicating 3 and containing content.signedData containing hashId indicating hash algorithm of the verification key of CERT_TS_A_AT and containing tbsData containing payload containing data containing protocolVersion indicating 3 and containing content.unsecuredData containing DENM payload and containing headerInfo containing psid indicating DENM AID value and containing generationTime indicating time within 2sec around the CUR_TIME and containing generationLocation and NOT containing other headers and containing signer containing certificate containing 1 item of type EtsiTs103097Certificate indicating CERT_TS_A_AT and containing signature containing ecdsaNistP256Signature containing rSig.x-only calculated over the MSG.content.signedData.tbsData using verification key of CERT_TS_A_AT then the IUT accepts the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_02_BV_***XX***
Summary	Check that IUT accepts a valid secured DENM message signed with certificate containing region restriction
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY AND ***X_PICS***

Expected behaviour

with
the IUT being in the 'authorized' state
and the IUT current time (CUR_TIME) is inside the time validity period of ***X_AT_CERTIFICATE***
and the IUT current position is inside the region restriction of ***X_AT_CERTIFICATE***
ensure that
when
the IUT is receiving a message of type EtsiTs103097Data (MSG)
indicating the message described in TP_SEC_ITSS_RCV_DENM_01_BV
and containing content.signedData
containing tbsData
containing headerInfo
containing generationLocation
indicating location inside the ***X_AT_CERTIFICATE*** region restriction
and containing signer
containing certificate
containing 1 item of type EtsiTs103097Certificate
indicating ***X_AT_CERTIFICATE***
containing toBeSigned.region
containing ***X_FIELD***
and containing signature
containing ecdsaNistP256Signature
containing rSig.x-only
calculated over the MSG.content.signedData.tbsData
using verification key of ***X_AT_CERTIFICATE***
then
the IUT accepts the SecuredMessage

Permutation Table

_XX	***X_FIELD***	***X_AT_CERTIFICATE***	***X_PICS***
01	circularRegion	CERT_TS_B_AT	PICS_SEC_CIRCULAR_REGION
02	rectangularRegion	CERT_TS_C_AT	PICS_SEC_RECTANGULAR_REGION
03	polygonalRegion	CERT_TS_D_AT	PICS_SEC_POLYGONAL_REGION
04	identifiedRegion	CERT_TS_E_AT	PICS_SEC_IDENTIFIED_REGION

TP Id	TP_SEC_ITSS_RCV_DENM_03_BV
Summary	Check that IUT accepts a valid secured DENM message signed using the brainpoolP256r1 algorithm
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_BRAINPOOL_P256R1
Expected behaviour	

With
the IUT being in the 'authorized' state
and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_B_AT
ensure that
when
the IUT is receiving a message of type EtsiTs103097Data (MSG)
indicating the message described in TP_SEC_ITSS_RCV_DENM_01_BV
and containing content.signedData
containing signer
containing certificate
containing 1 item of type EtsiTs103097Certificate
indicating CERT_TS_A_B_AT
containing toBeSigned.verifyKeyIndicator.verificationKey
containing ecdsaBrainpoolP256r1
and containing signature
containing ecdsaBrainpoolP256r1Signature
containing rSig.x-only
calculated over the MSG.content.signedData.tbsData
using verification key of CERT_TS_A_B_AT
then
the IUT accepts the SecuredMessage

TP Id	TP_SEC_ITSS_RCV_DENM_04_BV
Summary	Check that IUT accepts a valid secured DENM message signed using the brainpoolP384r1 algorithm
Reference	ETSI TS 103 097 [1], clause 7.1.1
PICS Selection	PICS_GN_SECURITY AND PICS_SEC_BRAINPOOL_P384R1
Expected behaviour	

With
the IUT being in the 'authorized' state
and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_B3_AT
ensure that
when
the IUT is receiving a message of type EtsiTs103097Data (MSG)
indicating the message described in TP_SEC_ITSS_RCV_DENM_01_BV
and containing content.signedData
containing signer
containing certificate
containing 1 item of type EtsiTs103097Certificate
indicating CERT_TS_A_B3_AT
containing toBeSigned. verifyKeyIndicator.verificationKey
containing ecdsaBrainpoolP384r1
and containing signature
containing ecdsaBrainpoolP384r1Signature
containing rSig.x-only
calculated over the MSG.content.signedData.tbsData
using verification key of CERT_TS_A_B3_AT
then
the IUT accepts the SecuredMessage

6.3.3.2 Check invalid HeaderInfo elements

TP Id	TP_SEC_ITSS_RCV_DENM_01_BO
Summary	Check that IUT discards a secured DENM if the HeaderInfo contains the header field an invalid Psid value
Reference	ETSI TS 103 097 [1], clause 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing psid **not** indicating DENM AID value then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_02_BO
Summary	Check that IUT discards a secured DENM if the HeaderInfo does not contain the header field generationLocation
Reference	ETSI TS 103 097 [1], clause 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing Psid indicating DENM AID value and **not** containing generationLocation then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_03_BO
Summary	Check that IUT discards a secured DENM if the HeaderInfo contains the header field expiryTime
Reference	ETSI TS 103 097 [1], clause 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing Psid indicating DENM AID value and containing expiryTime then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_04_BO
Summary	Check that IUT discards a secured DENM if the HeaderInfo contains the header field p2pcdLearningRequest
Reference	ETSI TS 103 097 [1], clause 5.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing Psid indicating DENM AID value and containing p2pcdLearningRequest then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_05_BO
Summary	Check that IUT discards a secured DENM if the HeaderInfo contains the header field missingCrlIdentifier
Reference	ETSI TS 103 097 [1], clause 5.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing Psid indicating DENM AID value and containing missingCrlIdentifier then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_06_BO
Summary	Check that IUT discards a secured DENM if the HeaderInfo contains the header field encryptionKey
Reference	ETSI TS 103 097 [1], clause 7.1.2
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
With the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing ToBeSignedData containing HeaderInfo containing Psid indicating DENM AID value and containing encryptionKey then the IUT discards the SecuredMessage	

6.3.3.3 Check invalid Signature elements

TP Id	TP_SEC_ITSS_RCV_DENM_07_BO
Summary	Check that IUT discards a secured DENM if the 'SignedData' contains an invalid signature algorithm
Reference	ETSI TS 103 097 [1], clause 6
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing Signature indicating invalid signature algorithm then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_08_BO
Summary	Check that IUT discards a secured DENM if the 'SignerIdentifier' contains an invalid choice
Reference	ETSI TS 103 097 [1], clause 6
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing SignedData containing SignerIdentifier indicating 'self' then the IUT discards the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_DENM_09_BO
Summary	Check that IUT discards a secured DENM if the Signature cannot be verified
Reference	ETSI TS 103 097 [1], clause 6
PICS Selection	PICS_GN_SECURITY
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time is inside the time validity period of CERT_TS_A_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data containing Signature indicating an altered value different from the one calculated using the signature calculation rules then the IUT discards the SecuredMessage	

6.4 Implicit certificate

TP Id	TP_SEC_ITSS_SND_IMPLICIT_CERT_01_BV
Summary	Check that IUT supports usage of implicit certificate for signing message
Reference	ETSI TS 103 097 [1], Clauses 5.2 & 7.1.1
PICS Selection	PICS_GN_SECURITY and (PICS_SEC_NIST_P256 or PICS_SEC_BRAINPOOL_P256R1)
Expected behaviour	
with the IUT is authorized with AT implicit certificate (X_CERTIFICATE) containing verifyKeyIndicator containing reconstructionValue containing R_KEY ensure that when the IUT is requested to send a secured CAM then the IUT sends a Ieee1609Dot2Data containing content containing signedData containing signer containing certificate containing 1 item of type EtsiTs103097Certificate indicating X_CERTIFICATE and containing signature containing ecdsaNistP256Signature virified with verification key reconstructed from reconstruction value of X_CERTIFICATE	

TP Id	TP_SEC_ITSS_RCV_IMPLICIT_CERT_01_BV
Summary	Check that IUT accepts a valid secured CAM message signed with a known implicit certificate
Reference	ETSI TS 103 097 [1] Clauses 5.2 & 7.1.1
PICS Selection	PICS_GN_SECURITY and (PICS_SEC_NIST_P256 or PICS_SEC_BRAINPOOL_P256R1)
Expected behaviour	
with the IUT being in the 'authorized' state and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_IMP_AT ensure that when the IUT is receiving a message of type EtsiTs103097Data (MSG) containing protocolVersion indicating 3 and containing content.signedData containing hashId indicating hash algorithm of the reconstruction key of CERT_TS_A_IMP_AT and containing tbsData and containing signer containing certificate containing 1 item of type EtsiTs103097Certificate indicating CERT_TS_A_IMP_AT and containing signature containing ecdsaNistP256Signature containing rSig.x-only calculated over the MSG.content.signedData.tbsData using verification key of CERT_TS_A_IMP_AT then the IUT accepts the SecuredMessage	

TP Id	TP_SEC_ITSS_RCV_IMPLICIT_CERT_02_BV
Summary	Check that IUT accepts a valid secured CAM message signed with an unknown implicit certificate
Reference	ETSI TS 103 097 [1] Clauses 5.2 & 7.1.1
PICS Selection	PICS_GN_SECURITY and (PICS_SEC_NIST_P256 or PICS_SEC_BRAINPOOL_P256R1)

Expected behaviour

```
with
    the IUT being in the 'authorized' state
    and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_F_AT_IMP
ensure that
   when
      the IUT is receiving a message of type EtsiTs103097Data (MSG)
         containing protocolVersion
            indicating 3
         and containing content.signedData
            containing hashId
               indicating hash algorithm of the reconstruction key of CERT_TS_F_AT_IMP
            and containing tbsData
         and containing signer
            containing certificate
               containing 1 item of type EtsiTs103097Certificate
                  indicating CERT_TS_F_AT_IMP
         and containing signature
            containing ecdsaNistP256Signature
               containing rSig.x-only
            calculated over the MSG.content.signedData.tbsData
               using verification key of CERT_TS_F_AT_IMP
then
      the IUT accepts the SecuredMessage
```

TP Id	TP_SEC_ITSS_RCV_IMPLICIT_CERT_01_BO
Summary	Check that IUT discards a valid secured CAM message signed with implicit certificate containing signature
Reference	ETSI TS 103 097 [1] Clauses 5.2 & 7.1.1
PICS Selection	PICS_GN_SECURITY and (PICS_SEC_NIST_P256 or PICS_SEC_BRAINPOOL_P256R1)

Expected behaviour

```
with
    the IUT being in the 'authorized' state
    and the IUT current time (CUR_TIME) is inside the time validity period of CERT_TS_A_IMP_AT_BO
ensure that
   when
      the IUT is receiving a message of type EtsiTs103097Data (MSG)
         containing protocolVersion
            indicating 3
         and containing content.signedData
            containing hashId
               indicating hash algorithm of the reconstruction key of CERT_TS_A_IMP_AT_BO
            and containing tbsData
         and containing signer
            containing certificate
               containing 1 item of type EtsiTs103097Certificate
                  indicating CERT_TS_A_IMP_AT_BO
         and containing signature
            containing ecdsaNistP256Signature
               containing rSig.x-only
            calculated over the MSG.content.signedData.tbsData
               using verification key of CERT_TS_A_IMP_AT_BO
then
      the IUT discards the SecuredMessage
```

Annex A (informative): Bibliography

- ETSI TS 102 894-2 (V1.2.1): "Intelligent Transport Systems (ITS); Users and applications requirements; Part 2: Applications and facilities layer common data dictionary".

History

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